

HCI Project Proposal: University Shuttle Tracking System

Submission Deadline: Wednesday, 10th June, 2026. 7am Prompt

System Description

The University Shuttle Tracking System is a web-based and mobile-friendly platform designed to help students track campus shuttle buses in real-time. The system provides information about shuttle locations, routes, estimated arrival times, and service updates. It aims to reduce waiting times, improve transportation planning, and enhance the overall commuting experience for students.

Intended Users

Primary Users: University students who use campus shuttle services.

Secondary Users: Shuttle drivers and University transport coordinators.

Problem Statements

1. **Students** need a way to **track shuttle locations in real-time** because **they often wait indefinitely at stops due to a lack of schedule certainty**.
2. **Students** need a way to **view integrated route maps and schedules** because **navigating between different shuttle lines is currently confusing and poorly documented**.
3. **Transport Coordinators** need a way to **monitor fleet performance and delays** because **they cannot optimize shuttle distribution or respond to bottlenecks without real-time data**.

Preliminary Usability Problems to Investigate

- Difficulty locating important information such as shuttle arrival times.
- Confusing navigation between routes, schedules, and tracking features.
- Poor visibility of notifications regarding delays or route changes.
- Difficulty understanding map symbols and shuttle locations.
- Excessive steps required to access frequently used features.
- Information overload caused by displaying too much transport information at once.
- Lack of accessibility support for users with visual impairments.

Expected Benefits

- Reduced waiting times for students.

- Improved transportation planning.
- Better communication between students and shuttle operators.
- Increased user satisfaction with campus transportation services.

Group Members

Name	Index Number
REDOFF OPPONG AGYEMANG	5230100081
LLYWELLYN GYAMERA KRA	5230100082
DAVID MPIANI OWUSU	5230100083
BETTY AGYENIM-BOATENG	5230100084
ALEX OPARE	5230100085
JOSEPH AANINNA	5230100087
JOSEPH MENSAH	5230100089
KELVIN ADJEI MENSAH	5230100090